

# Synthetic BOAMP Linkage Algorithm Benchmark Report

Generated from the corrected synthetic benchmark notebook

2026-08-04

## Technical Summary

Gradient boosting is the strongest current linker on the corrected synthetic benchmark. It reaches precision **0.719**, end-to-end recall **0.235**, and end-to-end pair-F1 **0.354** across the held-out evaluation grid.

The result is not mainly a classifier-only story. It is a pipeline story: the production candidate generator exposes only **10,333** of **32,404** in-scope truth links to scoring, a weighted blocking recall of **31.9%** and an unweighted world-average recall of **0.291**. This is why gradient boosting has fixed-candidate F1 **0.728** but end-to-end F1 only **0.354**.

Logistic regression recovers essentially the same number of true links as gradient boosting (7,704 versus 7,624), but accepts far more links (22,706 versus 10,610). The extra volume lowers precision from **0.719** to **0.339**. The current weighted composite and Fellegi-Sunter-style baseline remain useful comparators, but they are not the preferred scorers under this benchmark.

## Gradient Boosting Wins, Mostly By Reducing False Positives

The four algorithms were evaluated on the same held-out candidate-pair universe. Gradient boosting accepts fewer links than logistic regression and the composite baseline, while retaining nearly the same true-positive count as logistic regression. That makes it the best current choice for synthetic benchmark ranking and the least risky of the tested options for downstream survival-analysis pilots.

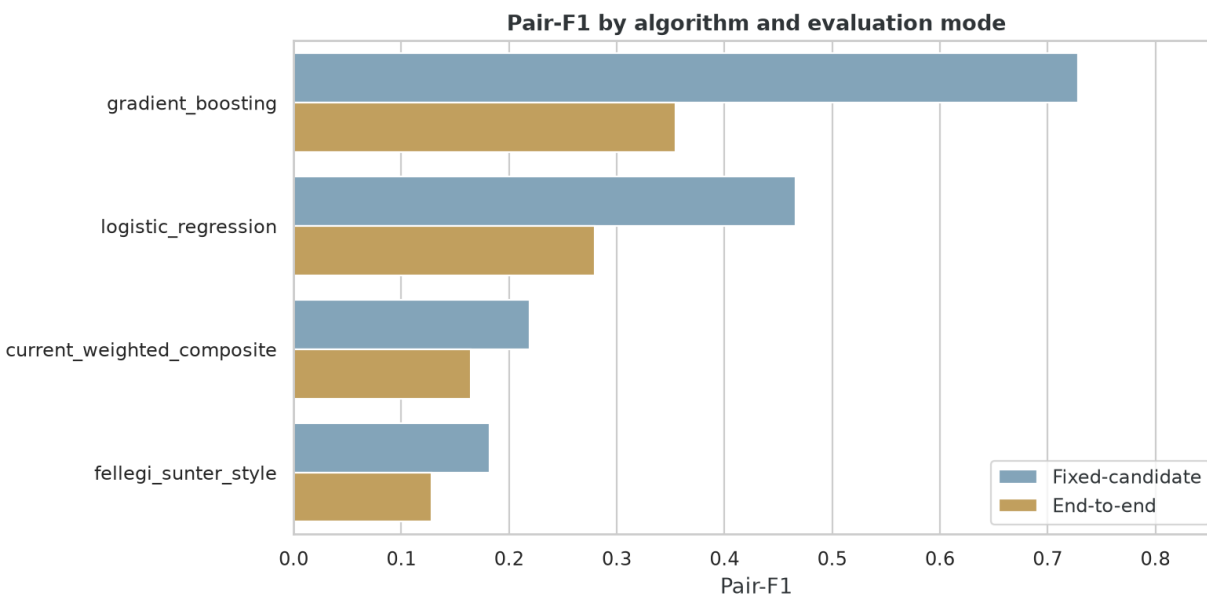


Figure 1: Pair-F1 by algorithm and evaluation mode. Fixed-candidate performance is much higher because unreachable truth links are excluded from that denominator.

Table 1: Aggregate held-out performance by algorithm.

| Algorithm                  | Predicted links | True positives | Precision | Fixed-candidate recall | End-to-end recall | End-to-end F1 |
|----------------------------|-----------------|----------------|-----------|------------------------|-------------------|---------------|
| Gradient boosting          | 10,610          | 7,624          | 0.719     | 0.738                  | 0.235             | 0.354         |
| Logistic regression        | 22,706          | 7,704          | 0.339     | 0.746                  | 0.238             | 0.280         |
| Current weighted composite | 56,103          | 7,288          | 0.130     | 0.705                  | 0.225             | 0.165         |
| Fellegi-Sunter style       | 41,909          | 4,752          | 0.113     | 0.460                  | 0.147             | 0.128         |

Compared with the current weighted composite, gradient boosting improves precision by **5.5x** and end-to-end F1 by **2.2x**.

## Score Diagnostics: ROC, Precision-Recall, Threshold F1, And GBM Loss

Because the synthetic benchmark has known truth, the algorithm evaluation can report full candidate-pair discrimination curves as well as rank-1 threshold behavior. These are benchmark-truth diagnostics only; they do not measure real BOAMP precision or recall.

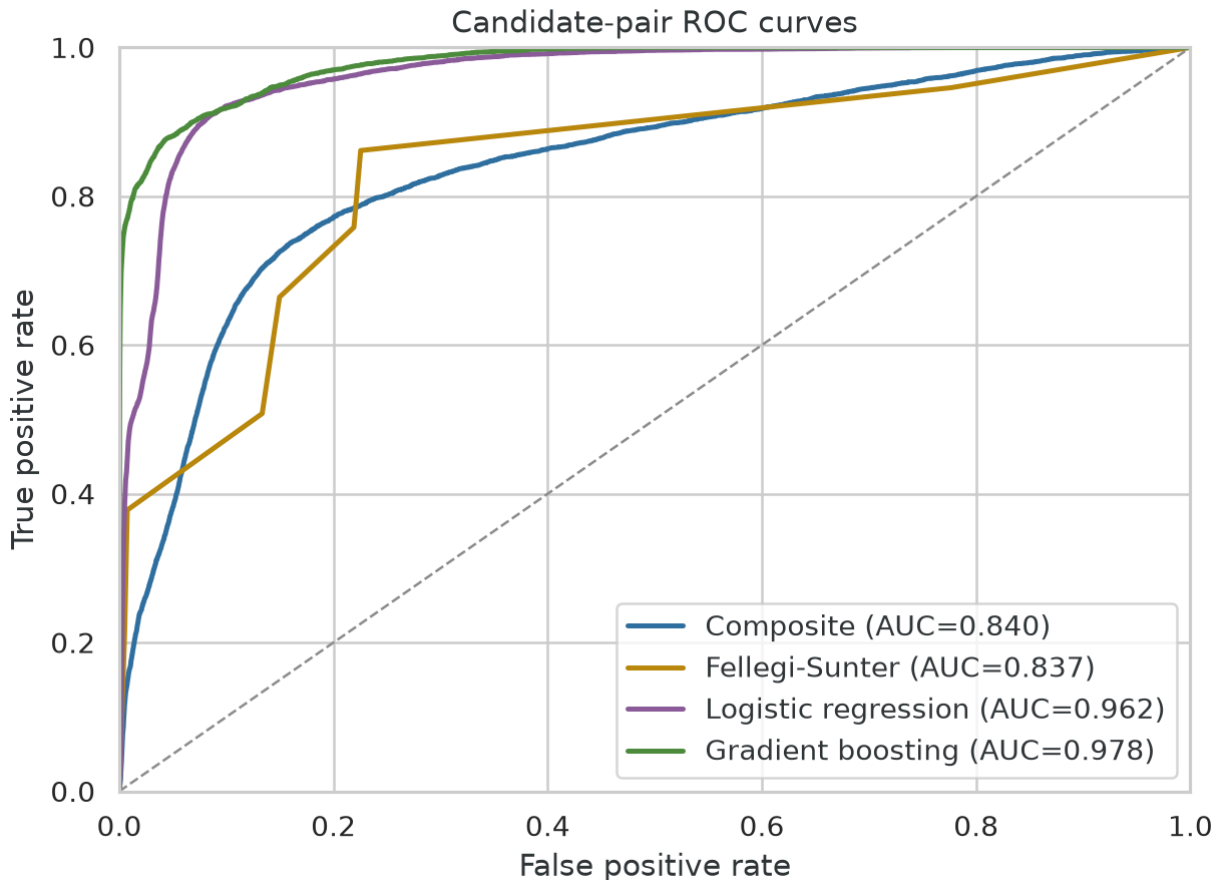


Figure 2: Candidate-pair ROC curves. Gradient boosting and logistic regression separate candidate pairs best by ROC-AUC.

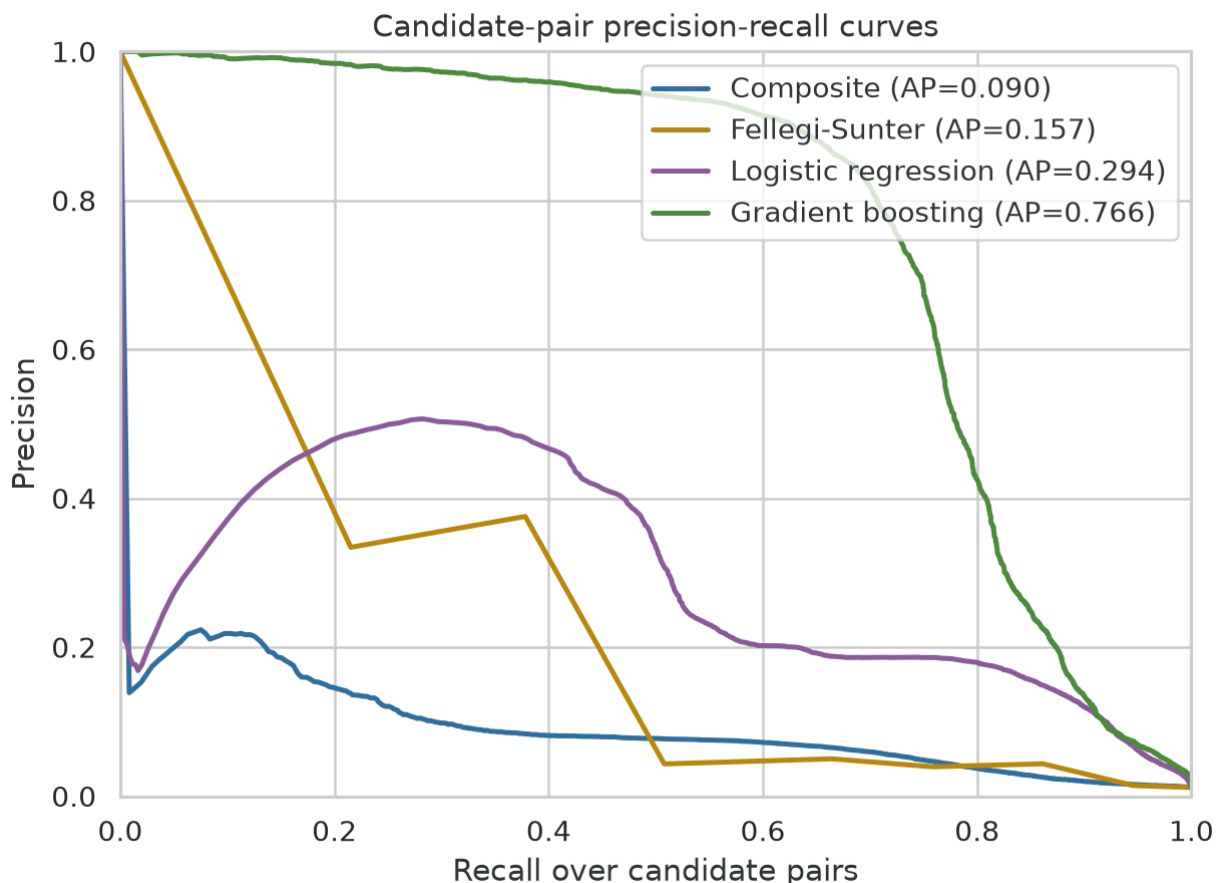


Figure 3: Candidate-pair precision-recall curves. Average precision is more diagnostic than ROC-AUC under low true-match prevalence.

Table 2: Curve diagnostics and frozen-threshold rank-1 performance.

| Algorithm                  | ROC-AUC | Average precision | Frozen precision | Frozen end-to-end recall | Frozen end-to-end F1 |
|----------------------------|---------|-------------------|------------------|--------------------------|----------------------|
| Current weighted composite | 0.840   | 0.090             | 0.130            | 0.225                    | 0.165                |
| Fellegi-Sunter style       | 0.837   | 0.157             | 0.113            | 0.147                    | 0.128                |
| Logistic regression        | 0.962   | 0.294             | 0.339            | 0.238                    | 0.280                |
| Gradient boosting          | 0.978   | 0.766             | 0.719            | 0.235                    | 0.354                |

The ROC curves show broad pair separability; the precision-recall curves are more informative under the low true-match prevalence of the candidate-pair universe. Gradient boosting has the strongest average precision and the highest frozen-threshold rank-1 F1.

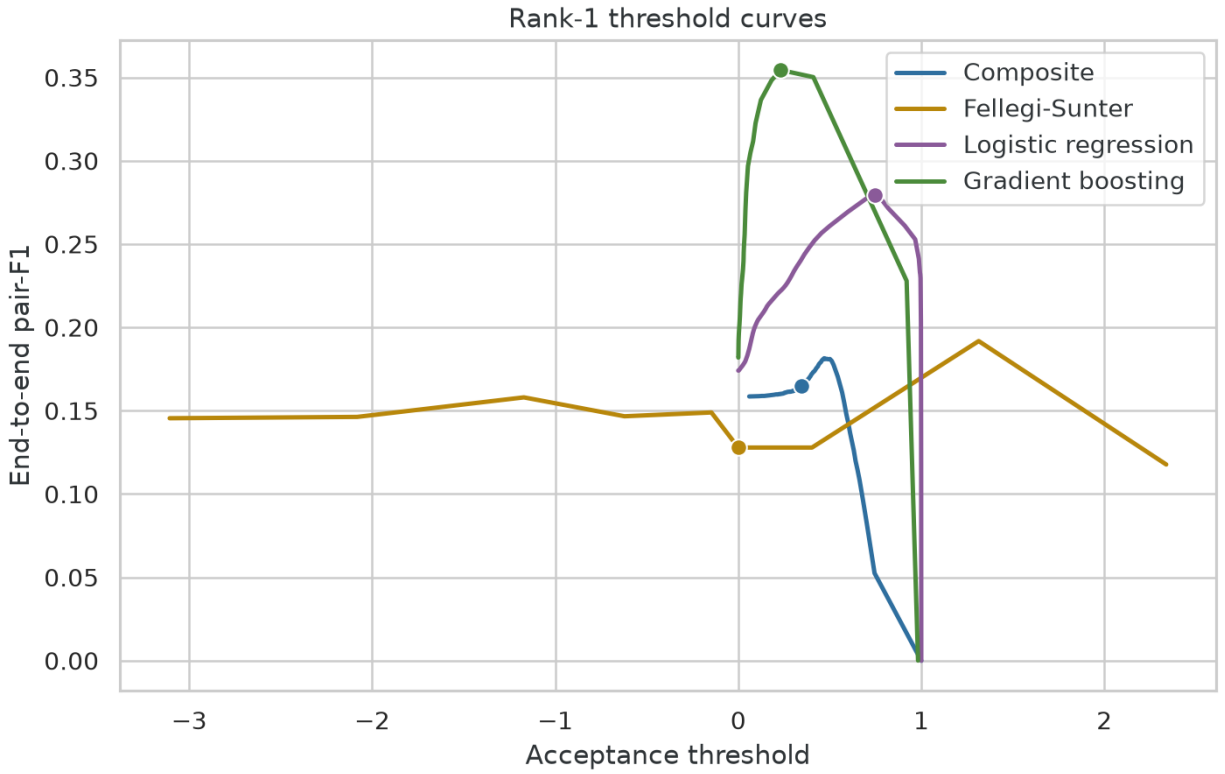


Figure 4: Rank-1 end-to-end F1 over acceptance thresholds. Points mark the frozen thresholds used in the benchmark.

The threshold curves show that the selected gradient-boosting threshold is at the held-out F1 peak in this grid, while logistic regression is close to its peak but at much lower precision.

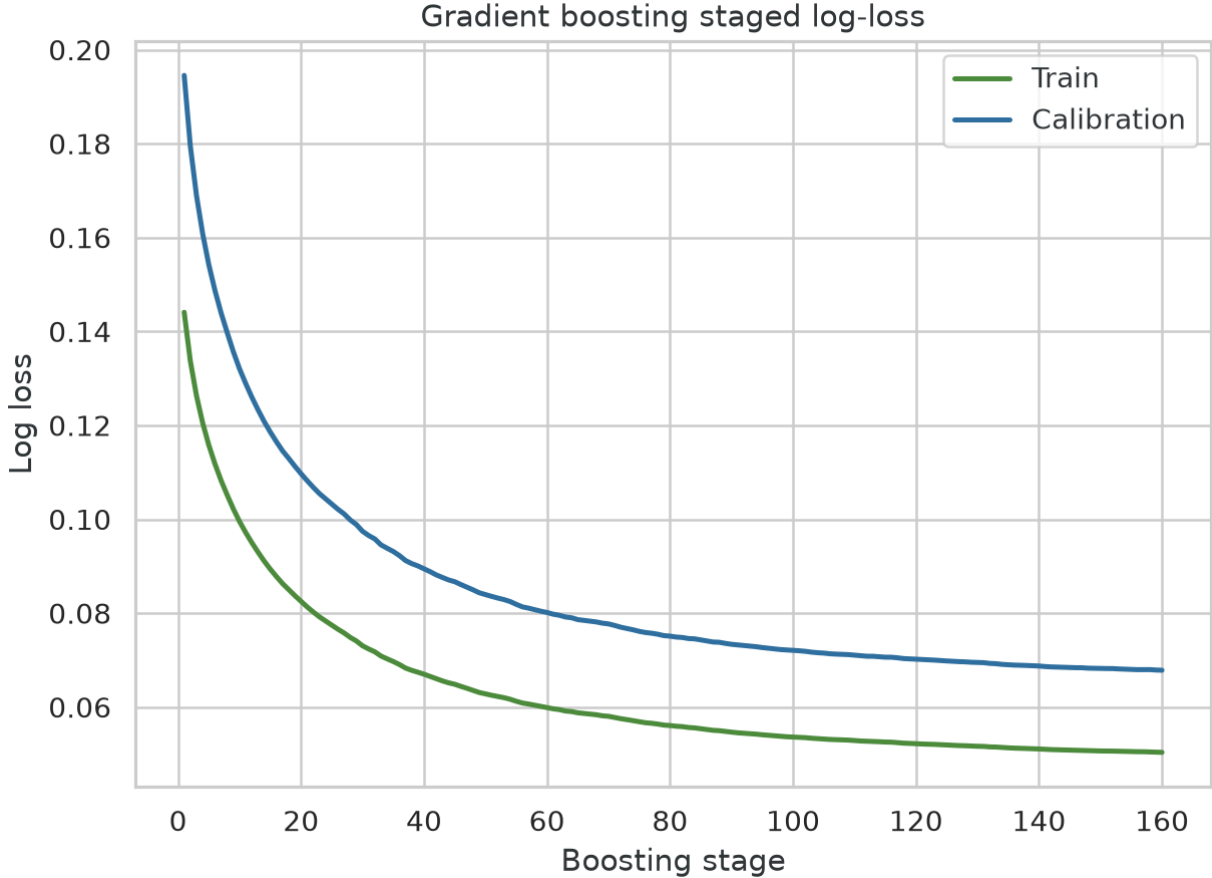


Figure 5: Gradient boosting staged log-loss on fit and calibration worlds.

The gradient-boosting staged log-loss falls smoothly on both the training and calibration worlds. The calibration curve remains above training loss, as expected, but does not show late-stage instability over the 160 boosting stages used by the frozen benchmark model.

## Candidate Generation Is The Binding Constraint

End-to-end performance is much lower than fixed-candidate performance because most truth links never enter the candidate set. The evaluation run scored **879,179** candidate pairs over **32,404** in-scope truth links, but only **10,333** truth links were reachable by the production candidate generator.

Table 3: Candidate-generation reachability by evaluation scenario.

| Scenario            | Worlds | Candidate pairs | Truth links | Reachable truth | Mean blocking recall | Min   | Max   |
|---------------------|--------|-----------------|-------------|-----------------|----------------------|-------|-------|
| central_provisional | 4      | 80,801          | 6,488       | 2,166           | 0.334                | 0.303 | 0.355 |
| difficult           | 4      | 177,492         | 3,823       | 881             | 0.234                | 0.191 | 0.286 |
| easier              | 4      | 145,729         | 11,707      | 4,475           | 0.383                | 0.345 | 0.410 |
| moderate            | 4      | 80,801          | 6,488       | 2,166           | 0.334                | 0.303 | 0.355 |
| stress              | 4      | 394,356         | 3,898       | 645             | 0.167                | 0.155 | 0.188 |

The stress scenario makes this clearest: it creates 386,410 candidate pairs but exposes only 648 of 4,030 truth links to scoring. No scoring model can recover links that were never generated. Therefore the next major gain should come from candidate generation, not only from another classifier.

## Scenario Results Show Robust Ranking But Different Difficulty

Gradient boosting ranks first in every evaluated scenario. The ranking is stable, but absolute performance changes sharply as the benchmark gets harder.

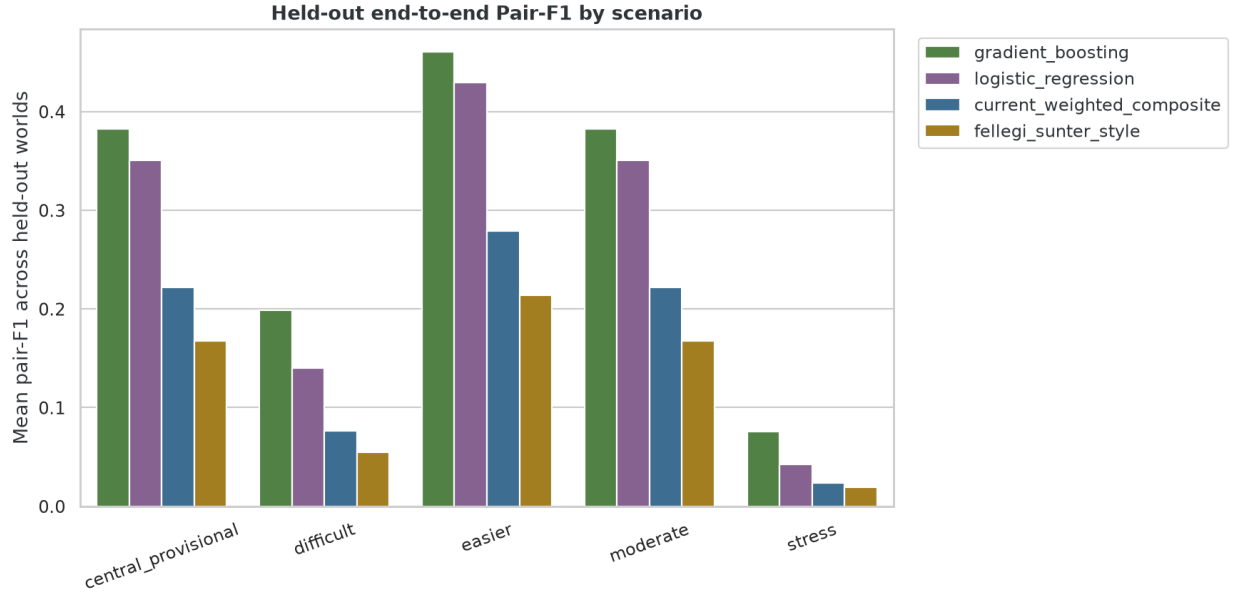


Figure 6: Held-out end-to-end pair-F1 by scenario. Gradient boosting leads throughout, but stress and difficult scenarios lower all algorithms.

Table 4: Mean held-out performance by scenario and algorithm.

| Scenario            | Algorithm                  | Mean F1 | Mean precision | Mean recall | SD F1 |
|---------------------|----------------------------|---------|----------------|-------------|-------|
| central_provisional | Gradient boosting          | 0.382   | 0.770          | 0.254       | 0.015 |
| central_provisional | Logistic regression        | 0.350   | 0.549          | 0.258       | 0.011 |
| central_provisional | Current weighted composite | 0.222   | 0.204          | 0.244       | 0.014 |
| central_provisional | Fellegi-Sunter style       | 0.168   | 0.181          | 0.158       | 0.012 |
| difficult           | Gradient boosting          | 0.199   | 0.475          | 0.126       | 0.023 |
| difficult           | Logistic regression        | 0.140   | 0.143          | 0.138       | 0.021 |
| difficult           | Current weighted composite | 0.077   | 0.054          | 0.133       | 0.024 |
| difficult           | Fellegi-Sunter style       | 0.055   | 0.043          | 0.077       | 0.019 |
| easier              | Gradient boosting          | 0.460   | 0.879          | 0.312       | 0.029 |
| easier              | Logistic regression        | 0.429   | 0.715          | 0.307       | 0.025 |
| easier              | Current weighted composite | 0.279   | 0.270          | 0.288       | 0.023 |
| easier              | Fellegi-Sunter style       | 0.214   | 0.242          | 0.192       | 0.024 |
| moderate            | Gradient boosting          | 0.382   | 0.770          | 0.254       | 0.015 |
| moderate            | Logistic regression        | 0.350   | 0.549          | 0.258       | 0.011 |
| moderate            | Current weighted composite | 0.222   | 0.204          | 0.244       | 0.014 |
| moderate            | Fellegi-Sunter style       | 0.168   | 0.181          | 0.158       | 0.012 |
| stress              | Gradient boosting          | 0.076   | 0.162          | 0.049       | 0.017 |
| stress              | Logistic regression        | 0.043   | 0.032          | 0.065       | 0.009 |
| stress              | Current weighted composite | 0.024   | 0.015          | 0.067       | 0.003 |
| stress              | Fellegi-Sunter style       | 0.019   | 0.012          | 0.042       | 0.005 |

The easier, central, and moderate settings support useful discrimination among algorithms. The difficult and stress settings are more diagnostic of failure modes. In stress, gradient boosting still leads, but end-to-end F1 falls to **0.122** because blocking recall and candidate ambiguity both worsen. Central and moderate are identical in this executed run, so they should not be interpreted as separate difficulty levels until their scenario definitions diverge.

## Mechanical Bias Remains Visible

The benchmark shows systematic performance differences by identifier quality, candidate count, schema, and CPV availability. These are expected linkage failure modes, but they matter because survival analysis could inherit them.

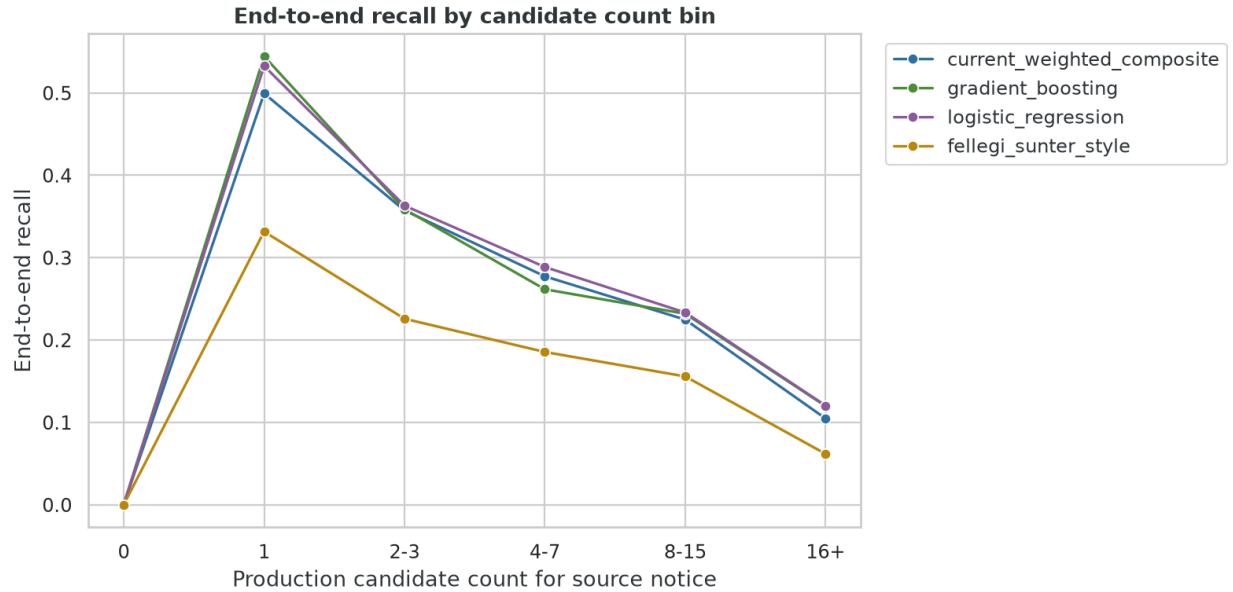


Figure 7: End-to-end recall by candidate count bin. Crowded candidate lists sharply reduce recall.

For gradient boosting, end-to-end recall is **0.544** when a source has exactly one candidate, but only **0.120** when it has sixteen or more candidates. This is the strongest bias diagnostic: crowded candidate environments are under-linked.

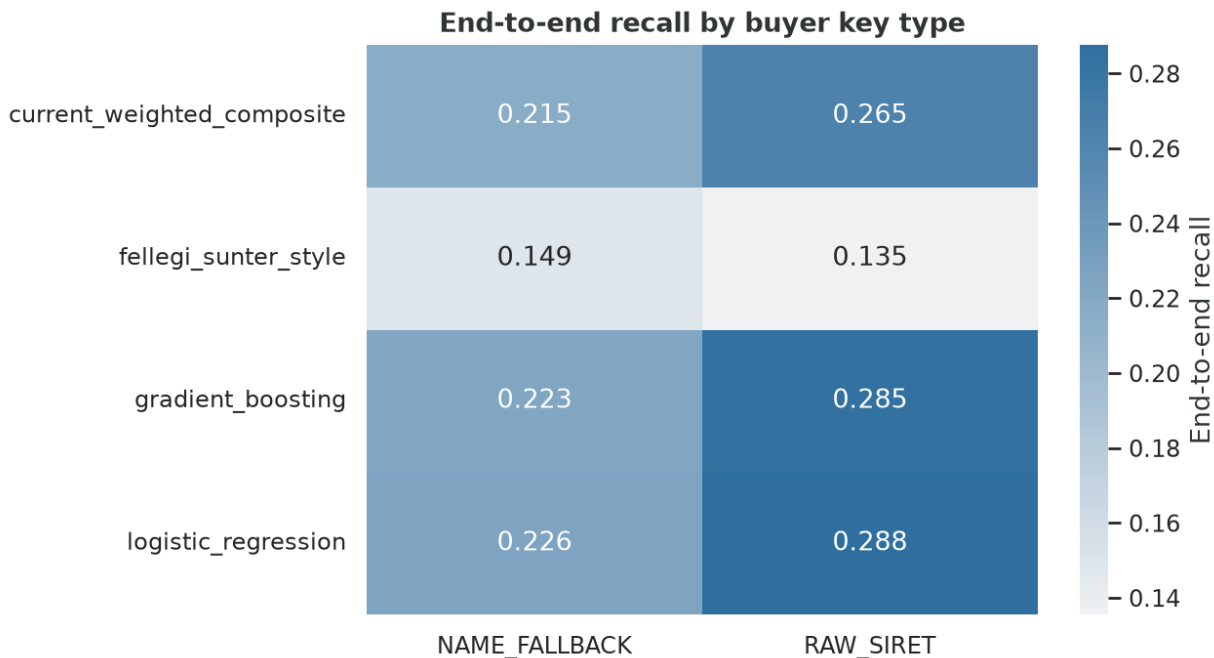


Figure 8: End-to-end recall by buyer key type. Cleaner identifiers remain easier to link.

Identifier quality also matters. Gradient boosting recall is **0.285** for RAW\_SIRET truth links versus **0.223** for NAME\_FALLBACK. CPV availability matters as well: recall is **0.251** when CPV is present and **0.099** when CPV is missing.

## Thresholds And Model Setup

The current composite and Fellegi-Sunter thresholds are frozen. Logistic regression and gradient boosting thresholds are selected on central calibration worlds 005 and 006 by F1 maximization. Training uses central worlds 001 through 004; evaluation uses worlds 007 through 010 across central provisional, easier, moderate, difficult, and stress scenarios.

Table 5: Algorithm score columns and thresholds used in the executed benchmark.

| Algorithm                  | Score column            | Threshold | Threshold source               |
|----------------------------|-------------------------|-----------|--------------------------------|
| Current weighted composite | composite_score         | 0.343167  | production primary threshold   |
| Fellegi-Sunter style       | fs_log_odds             | 0         | frozen zero log-odds threshold |
| Logistic regression        | logistic_score          | 0.745678  | calibration F1 maximum         |
| Gradient boosting          | gradient_boosting_score | 0.230964  | calibration F1 maximum         |

The supervised models use pair-level scoring features already produced by the project pipeline: text similarity, CPV similarity, time score, buyer reliability score, gap features, candidate count, schema family, buyer key type, CPV missingness, and duration missingness indicators. The report does not claim these features are unbiased; the bias diagnostics above show that several are mechanically important.

## Limitations, Robustness, And Interpretation

These results are valid for the current synthetic benchmark, not for real BOAMP ground truth. The synthetic truth labels are generated, so model ranking can still contain generator-mechanism bias. The



report supports the claim that gradient boosting is best among the four tested linkers on this synthetic benchmark; it does not prove real-world BOAMP linkage accuracy.

The aggregate metrics are corrected to use a global pair key containing scenario, world, and notice pair. This avoids collisions from repeated synthetic notice IDs across generated worlds. The notebook was rerun after that correction and all tables in this report use the corrected outputs.

The fixed-candidate view is useful for comparing scorers, but the end-to-end view is the decision-relevant pipeline metric. If downstream survival analysis needs low false positives, gradient boosting is the best candidate among the tested models. If it needs high recall, the current candidate generation must be widened or redesigned.

## Recommended Next Steps

1. Use gradient boosting as the primary supervised linker for the next synthetic benchmark round.
2. Keep logistic regression as the interpretable supervised baseline, and keep the current composite and Fellegi-Sunter style models as frozen reference baselines.
3. Improve candidate generation before treating linkage as production-ready. Test wider temporal windows, alternate buyer-name blocking, CPV fallback blocking, and high-activity buyer handling.
4. Add a manually labeled real BOAMP validation set before making claims about real BOAMP accuracy.
5. Run survival-analysis sensitivity with at least three linkage layers: current composite, gradient boosting, and a high-precision conservative subset.

## Further Questions

1. How much end-to-end recall is recovered by widening candidate generation without creating too many false-positive candidates?
2. Are high-activity buyers under-linked enough to bias survival estimates?
3. Can a conservative gradient-boosting threshold provide a high-precision survival layer while a broader threshold supports recall sensitivity?
4. Do manually labeled real BOAMP pairs preserve the same algorithm ranking observed in synthetic truth?

## Source Artifacts

- Benchmark version: `v0_4_population_alias_revision`
- Executed notebook: `notebooks/14_linkage_algorithm_benchmark_v0_4.ipynb`
- Summary table: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/summary_metrics.csv`
- Scenario table: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/scenario_summary_metrics.csv`
- Bias diagnostics: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/recall_slices.csv`
- ROC curve points: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/roc_curve_points.csv`

- Precision-recall curve points: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/precision_recall_curve_points.csv`
- Rank-1 F1 threshold curve: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/rank1_f1_threshold_curve.csv`
- GBM loss curve: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/gbm_loss_curve.csv`
- Diagnostic curve summary: `reports/tables/synthetic_benchmark/v0_4_population_alias_revision/linkage_algorithm_benchmark/diagnostic_curve_summary.csv`